

TASK- 5

END-TO-END ICSR PROCESSING

A Step-by-Step Guide to Processing One Adverse Event Report

What this document covers

The complete life cycle of one spontaneous adverse event report: intake and triage, the 4 minimum validity criteria, MedDRA coding, narrative writing, seriousness and expectedness classification, WHO-UMC causality assessment, the reporting decision, quality review, and final ICSR compilation.

This case is a fictional educational scenario created for pharmacovigilance internship practice. It is not a real patient record or real ICSR.

0. What Is an ICSR, and Why Does This Process Exist?

An **Individual Case Safety Report (ICSR)** is the standard document used worldwide to record one report of a suspected adverse reaction to a medicine, for a single patient. Every time a patient, doctor, pharmacist or company reports that someone may have had a bad reaction to a drug, that report is eventually turned into an ICSR and sent to the relevant health authority (in India, this flows through the Pharmacovigilance Programme of India; globally, authorities include the US FDA, the EU's EMA, and others).

In other terms: Think of an ICSR as a structured, standardized 'incident report' for a medicine side effect. Its job is to answer four questions in a consistent format so that safety scientists around the world can compare and combine thousands of these reports: Who was affected? What drug do we suspect? What happened to them? How bad was it, and was it already known?

A Drug Safety Associate (DSA) is the person who takes the messy, real-world version of a report — a phone call, an email, a form filled out by a nurse — and turns it into this structured ICSR, applying a consistent set of checks and assessments along the way. Those checks and assessments are the subject of this document.

The remainder of this document works through **one case, step by step, in the exact order a DSA would actually process it**, from the moment it lands in the inbox to the finished, submission-ready report.

1. The Case as Received (Raw Intake)

This is what actually arrives — before any assessment, coding or classification has been applied. Everything after this section is the DSA's work product, built from this raw information.

Field	As Reported
Source	Spontaneous report from a treating physician, via a hospital pharmacovigilance liaison
Patient	S.K., 39-year-old male
Relevant history	Major depressive disorder; chronic lower back pain
Suspect drug 1	Sertraline 50 mg orally, once daily, started 6 weeks prior
Suspect drug 2 (interacting)	Tramadol 50 mg orally, twice daily, added 2 days prior by a different prescriber
Reported event (free text)	"Agitation, confusion, tremor, sweating, rapid heartbeat and high temperature, starting a few hours after the new pain medicine was added."
Action taken	Both medicines stopped; patient admitted for observation and supportive care
Outcome at time of report	Improving; not yet fully recovered

STEP 1 — Initial Triage and Case Book-In

In other terms: This is the sorting step. Before doing any medical assessment, the DSA logs the report into the safety database and does a quick first-pass check: is this urgent, is it complete enough to work with, and who needs to see it first?

Actions performed at book-in:

- Assign a unique case identifier in the safety database.
- Record the date the report was first received (this date starts the regulatory reporting clock, described in Step 9).
- Capture the raw reporter information and free-text event description exactly as given, without editing or interpreting it yet.
- Do a first-pass scan for 'red flag' words (death, life-threatening, hospital, ICU, disability) that would mean the case needs to be fast-tracked to a senior reviewer.

Applied to this case: The free text mentions hospitalization/admission and high temperature with rapid heartbeat — both red-flag terms. This case is therefore flagged at book-in for priority handling, ahead of the formal seriousness assessment in Step 6.

STEP 2 — The Four Minimum Validity Criteria

In other terms: Not every report that arrives is actually usable. Before any further work happens, the DSA checks the report against four basic requirements. If even one is missing, the report is not a valid ICSR yet — it becomes an 'incomplete case' pending follow-up, rather than something that gets fully assessed and reported.

#	Criterion	What It Means	Met in This Case?
1	Identifiable patient	Enough detail to distinguish this patient from any other (initials, age/date of birth, or sex — a full legal name is not required).	Yes — 39-year-old male, initials S.K.
2	Identifiable reporter	Enough detail to know who is making the report and to follow up with them if needed (e.g. a named physician and a contactable facility).	Yes — treating physician, via hospital PV liaison
3	Suspect drug	At least one medicine is named as suspected of causing the event.	Yes — sertraline and tramadol both named
4	Adverse event	A specific reaction, symptom or clinical outcome is described — not just 'patient became unwell'.	Yes — described symptom cluster (agitation, tremor, sweating, tachycardia, fever)

Validity determination: VALID CASE. All four minimum criteria are met, so this report qualifies as a valid ICSR and proceeds to full processing. If any criterion had been missing (for example, no way to contact the reporter), the case would instead be logged as 'pending follow-up' and a follow-up request would be sent before further assessment.

STEP 3 — Seriousness-Based Triage / Prioritization

In other terms: Once a case is confirmed valid, it is prioritized for processing speed — similar to how a hospital emergency room triages patients. A case that looks potentially serious is worked on before a case that looks mild, even though the formal seriousness classification (Step 6) hasn't been finalized yet.

Applied to this case: Based on the book-in red flags (hospitalization, high fever, rapid heart rate), this case is assigned to the **high-priority processing queue**, ahead of routine (non-hospitalization) cases in the DSA's workload. This ensures a potentially serious case is not sitting unread behind lower-urgency reports.

STEP 4 — ADR Identification and MedDRA Coding

In other terms: MedDRA (the Medical Dictionary for Regulatory Activities) is a standardized medical vocabulary used across the pharmaceutical industry and regulators worldwide. Instead of leaving a reaction as free text ('felt shaky and hot'), the DSA translates it into one specific, standardized medical term from this dictionary. This is what lets a regulator combine and compare reports from different countries, languages and reporters.

MedDRA has a five-level hierarchy, from broadest to most specific:

Level	Name	Role	Example for This Case
1 (broadest)	System Organ Class (SOC)	Body system affected	Nervous system disorders
2	High Level Group Term (HLGT)	Broad grouping within the system	Neurological disorders NEC
3	High Level Term (HLT)	Narrower grouping	Serotonergic syndromes
4	Preferred Term (PT)	The standard term used for coding and reporting	Serotonin syndrome
5 (most specific)	Lowest Level Term (LLT)	The exact wording closest to what was reported	Serotonin syndrome

How the coding decision was made: The reporter described several separate-sounding symptoms — agitation, confusion, tremor, sweating, rapid heartbeat, high temperature. Rather than coding each symptom as its own separate MedDRA term, the DSA recognizes that this specific combination, occurring after a serotonergic drug was added, is a single recognized clinical syndrome. Coding it as one precise PT (Serotonin syndrome) is more accurate and more useful for safety monitoring than five unrelated-looking symptom codes.

In other terms: This step is similar to a doctor writing a proper diagnosis on a discharge summary instead of just listing every symptom the patient mentioned. Good coding requires enough clinical knowledge to recognize the pattern, not just data-entry skill.

Suspect medicine(s) identified for this ADR: Sertraline (pre-existing) and Tramadol (recently added) — assessed together, because the reaction is attributable to their combination rather than to either drug individually.

STEP 5 — Case Narrative Writing

In other terms: The narrative is a short, plain-language clinical story of the case — written so that a doctor reading it six months later, with no other context, can understand exactly what happened, in what order, and why. It is the single most important piece of free text in the whole ICSR, because every coded field (drug, event, seriousness, outcome) should be traceable back to something stated in the narrative.

Principles used when writing a good narrative:

- Present events in **chronological order** (drug started → drug added → symptoms began → action taken → outcome).
- State the **patient's relevant history** only if it is relevant to interpreting the event.
- Describe what actually happened in **plain clinical language**, avoiding unexplained abbreviations.
- Include **dechallenge** (what happened after stopping the drug) and **rechallenge** (what happened if the drug was restarted), if known — these matter for causality assessment in Step 8.
- End with the **outcome** at the time of reporting.

Narrative for this case:

S.K. is a 39-year-old male with a six-week history of sertraline 50 mg once daily for major depressive disorder. For chronic lower back pain, a second prescriber adds tramadol 50 mg twice daily, without cross-checking the patient's existing medication list. Within several hours of the second tramadol dose, the patient develops agitation and confusion, followed by tremor, profuse sweating, a rapid heart rate and a measured temperature of 38.9°C. He is brought to the emergency department, where the treating team recognizes the combination of neuro-excitatory, neuromuscular and autonomic findings as serotonin syndrome. Both sertraline and tramadol are discontinued immediately, and he is admitted for intravenous fluids, cooling measures and close monitoring. His agitation and vital signs gradually improve over the following 48 hours.

STEP 6 — Seriousness Assessment (ICH E2A Criteria)

In other terms: 'Serious' in pharmacovigilance is a specific regulatory label, not a judgement of how unpleasant the symptom felt. A reaction is 'serious' only if it meets one of six defined outcomes below — it has nothing to do with pain level or inconvenience on its own.

Criterion	Definition
Death	The adverse event results in the patient's death.
Life-threatening	The patient was, in the view of the reporter, at immediate risk of death from the event as it occurred.
Hospitalization	The event requires inpatient hospitalization or prolongs an existing hospitalization.
Disability / Incapacity	The event results in persistent or significant disability or incapacity.
Congenital anomaly	The event is a congenital anomaly or birth defect in the offspring of an exposed patient.
Other medically important event	Does not meet the criteria above but may jeopardize the patient or require intervention to prevent one of the outcomes listed, based on medical judgement.

Applying each criterion to this case:

Criterion	Met?	Case Evidence
Death	No	Patient improving at time of reporting.
Life-threatening	Yes	Autonomic instability (hyperthermia, tachycardia) placed the patient at immediate risk without prompt intervention.
Hospitalization	Yes	Admitted for IV fluids, cooling measures and monitoring.
Disability/Incapacity	Not assessed	Life-threatening/hospitalization criteria already satisfied — one 'Yes' is enough to classify the whole case as serious.
Congenital anomaly	No	Not applicable to this case.
Other medically important	Not assessed	Criteria above already satisfied.

Seriousness Classification: **SERIOUS**

Only one criterion needs to be met for a case to be classified as serious. Here, two are met (life-threatening and hospitalization), which reinforces the classification but does not change it.

STEP 7 — Expectedness Assessment

In other terms: 'Expected' does not mean common, mild, or medically obvious — it means one specific thing: is this reaction already written down in the drug's official reference safety information (the approved product label or, within a company, its Company Core Data Sheet)? This is a document-comparison exercise, not a clinical opinion.

The interaction between serotonergic antidepressants (such as sertraline) and tramadol, leading to serotonin syndrome, is a well-documented, labeled drug-interaction warning for both agents in their reference safety information, including guidance to avoid or closely monitor the combination.

Expectedness Classification: EXPECTED

Why this matters: As shown in Step 9, a case's regulatory reporting speed depends on the combination of seriousness AND expectedness together — not on either one alone.

STEP 8 — Causality Assessment (WHO-UMC Scale)

In other terms: Just because an event happened after a drug was taken does not prove the drug caused it. Causality assessment is the structured process of judging how likely it is that the drug was actually responsible. The WHO-UMC scale is one of the most widely used systems for this, with six defined categories (Certain, Probable/Likely, Possible, Unlikely, Conditional/Unclassified, Unassessable/Unclassifiable).

Assessing this case against the WHO-UMC criteria:

Criterion	WHO-UMC Requirement	Case Evidence	Met?
Time relationship	Reasonable time relationship to drug intake.	Onset within hours of the interacting drug (tramadol) being added.	Yes
Alternative causes	Unlikely to be attributed to concurrent disease or another drug.	No infection, primary neurological or endocrine cause identified; symptom cluster specifically fits a drug interaction.	Yes
Dechallenge	Clinically reasonable response to withdrawal.	Symptoms improved progressively after both drugs were stopped and supportive care given.	Yes
Rechallenge	Not required for Probable/Likely.	No rechallenge performed; would not be clinically appropriate.	N/A
Event definitive	Consistent with a recognized reaction pattern.	Symptom cluster matches the well-described serotonergic toxicity syndrome for this drug combination.	Yes

WHO-UMC Causality Classification: Probable / Likely

The reaction has a reasonable, drug-interaction-consistent time relationship, no credible competing cause, and a clinically reasonable dechallenge. Rechallenge information is absent, which keeps the case below Certain, but rechallenge is not required for a Probable/Likely classification — and deliberately re-exposing the patient purely to confirm causality would not be appropriate here.

STEP 9 — Reporting Decision and Regulatory Timeline

In other terms: This is where seriousness and expectedness are combined to decide how fast the case legally has to be sent to health authorities. Regulators generally require the fastest turnaround only for cases that are both serious AND unexpected — because that combination represents genuinely new safety information that needs urgent attention. Cases that are serious-but-expected, or non-serious, follow slower, routine timelines (often bundled into periodic safety reports).

Seriousness	Expectedness	Typical Reporting Speed
Serious	Unexpected	Expedited (commonly a 15-calendar-day clock from first knowledge of the case)
Serious	Expected	Routine timeline; included in periodic safety reporting
Non-serious	Unexpected	Routine timeline; still tracked for emerging-signal monitoring
Non-serious	Expected	Routine timeline; lowest processing priority

Applying this to the case:

This case is **Serious + Expected**. It is reportable — it meets the minimum validity criteria and involves at least a Possible causal relationship — but because it is Expected, it does not meet the expedited (fastest-timeline) criterion on expectedness grounds alone. It follows the routine regulatory reporting timeline.

Internal prioritization is separate from the regulatory clock. Even though the regulatory timeline is routine, this case is still flagged internally as high-priority for medical review, because it is life-threatening and stems from a preventable drug-drug interaction that a prescriber-alerting system might have caught. A DSA should remember: the regulatory deadline and a company's own internal urgency flagging are two different, independent decisions.

STEP 10 — Quality Review / Peer Check

In other terms: Before a case is finalized, a second, more senior reviewer typically checks the DSA's work — this is standard practice in almost every regulated safety database, similar to a second signature on a lab result.

Typical quality-check questions asked at this stage:

- Does the MedDRA term actually match what the narrative describes?
- Is every seriousness criterion marked 'Yes' actually supported by something stated in the narrative?
- Is the expectedness conclusion based on the current version of the reference safety information, not an outdated one?
- Does the causality reasoning follow logically from the documented evidence, without inventing details not in the source report?
- Are all E2B fields in the final ICSR internally consistent with each other?

Applied to this case: On review, all fields (seriousness = Serious, expectedness = Expected, causality = Probable/Likely) are each traceable to a specific statement in the narrative, and the case passes quality review without further changes.

STEP 11 — Final ICSR Compilation

In other terms: This is the finished product — everything from the steps above, consolidated into the standard ICSR field layout used for actual regulatory submission (following the international ICH E2B structure: Case, Patient, Drug and Reaction sections).

ICSR Field	Entry
Case identifier	Fictional training case — Final Project, Training Edition
Report type	Spontaneous report, initial
Reporter qualification	Physician (qualified healthcare professional)
Patient (initials, age, sex)	S.K., 39 years, male
Relevant medical history	Major depressive disorder; chronic lower back pain
Suspect drug 1	Sertraline 50 mg, oral, once daily
Suspect drug 2	Tramadol 50 mg, oral, twice daily
Reaction (MedDRA PT)	Serotonin syndrome
Reaction onset	Hours after tramadol was added to existing sertraline therapy
Action taken with drug(s)	Both drugs withdrawn
Outcome	Recovering / improving at time of report
Seriousness	Serious — life-threatening, hospitalization

Expectedness	Expected — labeled drug interaction
Causality (WHO-UMC)	Probable / Likely
Regulatory reporting timeline	Routine (non-expedited); internally flagged high-priority for medical review
Quality review status	Passed — all fields traceable to narrative

STEP 12 — Follow-Up Management

In other terms: Many cases are not 'closed' after the first report — the patient may still be recovering, and lab results or a discharge summary may arrive later. A DSA tracks these open items and issues a follow-up ICSR when new information arrives.

Applied to this case: The patient is reported as 'improving but not yet fully recovered.' This case is left open pending a follow-up report confirming full recovery, discharge, and — if available — final vital sign and laboratory values from the admission, which would be added to a follow-up ICSR rather than the initial report.

13. Glossary of Key Terms

Term	Plain-English Meaning
ICSR	Individual Case Safety Report — the standard structured document for one adverse event report.
Spontaneous report	A report that comes in unprompted, e.g. from a doctor or patient, rather than from a clinical trial.
MedDRA	A standardized international medical dictionary used to code adverse events consistently.
Preferred Term (PT)	The standard MedDRA term used to label a specific adverse event for coding and reporting.
Causality assessment	The structured judgement of how likely it is that a drug actually caused a reported event.
WHO-UMC scale	A widely used causality assessment system with six categories, from Certain to Unassessable.
Seriousness	A specific regulatory classification (met if death, life-threatening, hospitalization, disability, birth defect, or another medically important outcome occurs) — not the same as 'severe' symptoms.
Expectedness	Whether a reaction is already listed in a drug's approved reference safety information (label/CCDS).

Expedited reporting	The fastest regulatory reporting timeline, generally reserved for cases that are both serious and unexpected.
Dechallenge	What happens to the reaction after the suspected drug is stopped.
Rechallenge	What happens to the reaction if the suspected drug is restarted.
Reference safety information	The current official document (label/CCDS) listing a drug's known risks, used to judge expectedness.
Follow-up report	An updated ICSR issued when new information about a previously reported case becomes available.

14. Learning Points

- Validity comes before assessment.** A case is not medically assessed at all until it passes the four minimum validity criteria — patient, reporter, drug, event.
- Triage runs on early red flags, not final conclusions.** A case can be fast-tracked at intake based on rough signals, well before the formal seriousness assessment is finished.
- Coding requires clinical judgement, not just data-entry.** Recognizing that several symptoms form one syndrome, rather than coding each symptom separately, is a skill built on medical pattern recognition.
- Seriousness, expectedness and causality are three separate questions.** Each has its own criteria and its own answer; none of them can be inferred from the others.
- The reporting decision is a combination, not a single field.** It is the pairing of seriousness and expectedness together that determines the regulatory clock — not either one in isolation.
- Internal urgency and regulatory deadline are not the same thing.** A company can — and often should — treat a case as high-priority internally even when it does not qualify for the fastest external regulatory timeline.
- Every field in the final ICSR should trace back to the narrative.** If a reviewer cannot find support for a coded field in the case narrative, that field needs to be re-examined.

15. Final Conclusion

This training walkthrough followed one case through the entire ICSR life cycle used by a Drug Safety Associate: initial triage and book-in, the four minimum validity criteria, seriousness-based prioritization, ADR identification and MedDRA coding, narrative writing, seriousness assessment under ICH E2A, expectedness assessment against reference safety information, WHO-UMC causality assessment, the combined reporting decision, quality review, final ICSR compilation, and follow-up management. Together, these steps represent the full, real-world workflow behind a single adverse event report — the same workflow a DSA is expected to apply consistently across every case they process.

Disclaimer: This case is a fictional training example. It is not a real patient record, a real ICSR, or clinical advice.

References

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